LEVEL UP ACADEMY

Class 10 | CBSE | Science

Drill: Assertion–Reason & Case-Based Questions (CBSE)

Chapter / Topic: Mixed chapters · Week 9 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Part A: for each Assertion–Reason pair choose (a) both A and R are true and R explains A; (b) both true but R does not explain A; (c) A true, R false; (d) A false, R true.
2. Tip: first decide whether A is true, then whether R is true. Only then ask: does R explain A?
3. Part B: read each passage carefully. Answer in full sentences, with units and balanced equations where needed.

PART A: ASSERTION–REASON

Q1. **A:** The colour of copper sulphate solution changes when a zinc rod is dipped in it.
R: Zinc is more reactive than copper.

Q2. **A:** Calcium carbonate decomposes on heating.
R: Decomposition reactions are generally exothermic.

Q3. **A:** Pure water is neutral.
R: Pure water has equal concentrations of H^+ and OH^- ions.

Q4. **A:** Sodium is stored under kerosene.
R: Sodium is a non-metal.

Q5. **A:** Ionic compounds have high melting points.
R: Ionic compounds are soluble in water.

Q6. **A:** Diamond is a good conductor of electricity.
R: In diamond, each carbon atom is bonded to four other carbon atoms.

Q7. **A:** The walls of capillaries are only one cell thick.
R: Materials are exchanged between the blood and the surrounding cells through the capillaries.

Q8. **A:** A deficiency of insulin causes diabetes.
R: Insulin is secreted by the thyroid gland.

Q9. **A:** In human beings, the sex of the child is decided by the father.
R: Half the sperm carry an X chromosome and half carry a Y chromosome.

Q10. **A:** A concave lens always forms a virtual image of a real object.
R: The power of a concave lens is negative.

Q11. **A:** The resistance of a wire decreases when its length is increased.
R: The resistance of a wire is directly proportional to its length.

Q12. **A:** Plants are called producers.
R: Plants release oxygen during photosynthesis.

Q13. **A:** Copper reacts with dilute hydrochloric acid to give hydrogen gas.
R: Copper is less reactive than hydrogen.

Q14. **A:** Bile juice does not contain any enzyme.
R: Bile is produced by the liver.

PART B: CASE-BASED QUESTIONS

Q15. Ethanol is used in medicines, as a solvent and as a fuel. Even small amounts of methanol are poisonous: in the liver it is oxidised to methanal, which can damage the optic nerve and cause blindness. To stop industrial ethanol from being misused for drinking, poisonous substances like methanol are added to it, along with a blue dye. This is called denatured alcohol. **[4]**

(i) What is denatured alcohol? **[1]**

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(ii) Why is methanol dangerous to drink? **[1]**

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(iii) Write the equation for the reaction of ethanol with sodium. How would you test the gas given off? **[2]**

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Q16. In tissue culture, new plants are grown by taking a few cells or a small piece of tissue from the growing tip of a plant. The tissue is placed in an artificial medium, where it divides rapidly to form a small mass of cells called a callus. The callus is then moved to another medium containing hormones for growth and differentiation. The plantlets are finally placed in soil. This method is widely used for ornamental plants. **[4]**

(i) What is a callus? **[1]**

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(ii) Is tissue culture a sexual or an asexual method of reproduction? **[1]**

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(iii) Give two advantages of tissue culture. **[2]**

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Q17. The resistivity (in Ω m, at 20 °C) of some materials is: silver 1.60×10^{-8}, copper 1.62×10^{-8}, aluminium 2.63×10^{-8}, tungsten 5.20×10^{-8} and nichrome 100×10^{-6}. Glass has a resistivity of 10^{10} to 10^{14} Ω m. **[4]**

(i) Which of these metals is the best conductor of electricity? **[1]**

(ii) Which material in the list is an insulator? **[1]**

(iii) Why is nichrome used in heating elements, and copper in connecting wires? **[2]**

Q18. A student traces the path of a ray of light through a rectangular glass slab. The ray bends towards the normal when it enters the glass, and away from the normal when it comes out. The emergent ray is parallel to the incident ray but shifted sideways. This sideways shift is called lateral displacement. **[4]**

(i) Why does the ray bend towards the normal when it enters the glass? **[1]**

(ii) Why is the emergent ray parallel to the incident ray? **[1]**

(iii) How would the lateral displacement change if a thicker slab were used? What does the refractive index of glass (1.5) tell us? **[2]**

~ Check your answers with the key and add every mistake to your error log! ~